Multimedia Signal Processing - 2nd Module

February 3, 2023

1. **Question (7 pts)**
   Consider two sources X and Y that emit symbols according to the following joint p.m.f.

	y_1	y_2	y_3	y_4
x_1	1/8	0	0	1/8
x_2	0	0	1/4	0
x_3	0	1/8	1/8	0
x_4	0	0	?	1/4

   (a) Fill-in the joint p.m.f. table missing values (i.e., the question mark) in order to make it a proper and valid p.m.f.

   (b) Compute the probability mass functions of source X and source Y, separately.

   (c) Compute the entropy of both sources, i.e., $H(X)$ and $H(Y)$.

   (d) Compute the joint entropy $H(X, Y)$.

   (e) Compute the mutual information $I(X, Y)$.

   (f) Design the Huffman code to encode single symbols of X and compare the average code length to $H(X)$.

   (g) Design the Huffman code to jointly encode symbols of X and Y and compare the average code length to $H(X, Y)$.

2. **Question (4 pts)** Consider the problem of encoding an audio signal using a perceptual audio coding technique.

   (a) Report the main block diagram of a perceptual audio encoder.

   (b) Describe the concept of absolute threshold of hearing, sketching it with clear labels on the axes.

   (c) Highlight which is the role of the absolute threshold of hearing within the diagram.

3. **Question (4 pts)** Consider the process of encoding an image using the JPEG standard.

   (a) Report the block diagram of the encoder, clearly explaining the input and output of each block.

   (b) With reference to the reported diagram, explain what corresponds to the encoded signal.

   (c) Provide the definition of a quantization matrix (or quantization table), explaining its role in the JPEG standard.

4. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Consider the following code containing three codewords: $C_1 = 0$, $C_2 = 01$, $C_3 = 10$. This is a one-to-one code.
   - [T] [F] Consider a discrete memoryless source X with the following alphabet $\mathcal{A} = \{\alpha, \beta, \gamma, \delta\}$. This source can have entropy $H(X) = 1$.
   - [T] [F] Lloyd-Max quantizer is equivalent to a uniform scalar quantizer for uniformly distributed sources.
   - [T] [F] In a video coding scheme, motion vectors must be computed for each frame of the sequence.
   - [T] [F] Optimal bit allocation can be always achieved for transform coding schemes.
   - [T] [F] The use of KLT rather than DCT in a transform coding scheme leads to a possible increase of header information in the encoded file.
   - [T] [F] It is not worth using DPCM schemes if the variance of the prediction residual is lower than the variance of the input signal to be encoded.
   - [T] [F] Considering speech coding, the error weighting function of an analysis-by-synthesis linear predictive coder typically attenuates frequency bands around voice formants.

5. **Question (Matlab - 10pts)**

   Load the build-in 'pears.png' image file using the command `img = imread('pears.png')`.

   (a) Crop the top-left 256×256 pixels of the image and display it in full color (use the cropped version hereinafter).

   (b) Convert to grayscale and display the image again. Feel free to chose the RGB-to-gray conversion you want.

   (c) Quantize the grayscale image using a uniform scalar quantizer with $\{2, 3, 4, \ldots, 8\}$ bit/symbol

   (d) Plot the SNR vs. rate (use entropy as measure of rate).

   (e) Apply a 2D-DCT block-wise transform encoder: use 8×8 pixels blocks, 2D-DCT transform, and a uniform scalar quantizer with 4 bit/symbol.

   (f) Considering this last coding technique, compute the SNR and the rate (use entropy of the encoded signal as measure of rate).

   (g) How can you use the computed data to tell which one of the implemented coding scheme is better than the others?